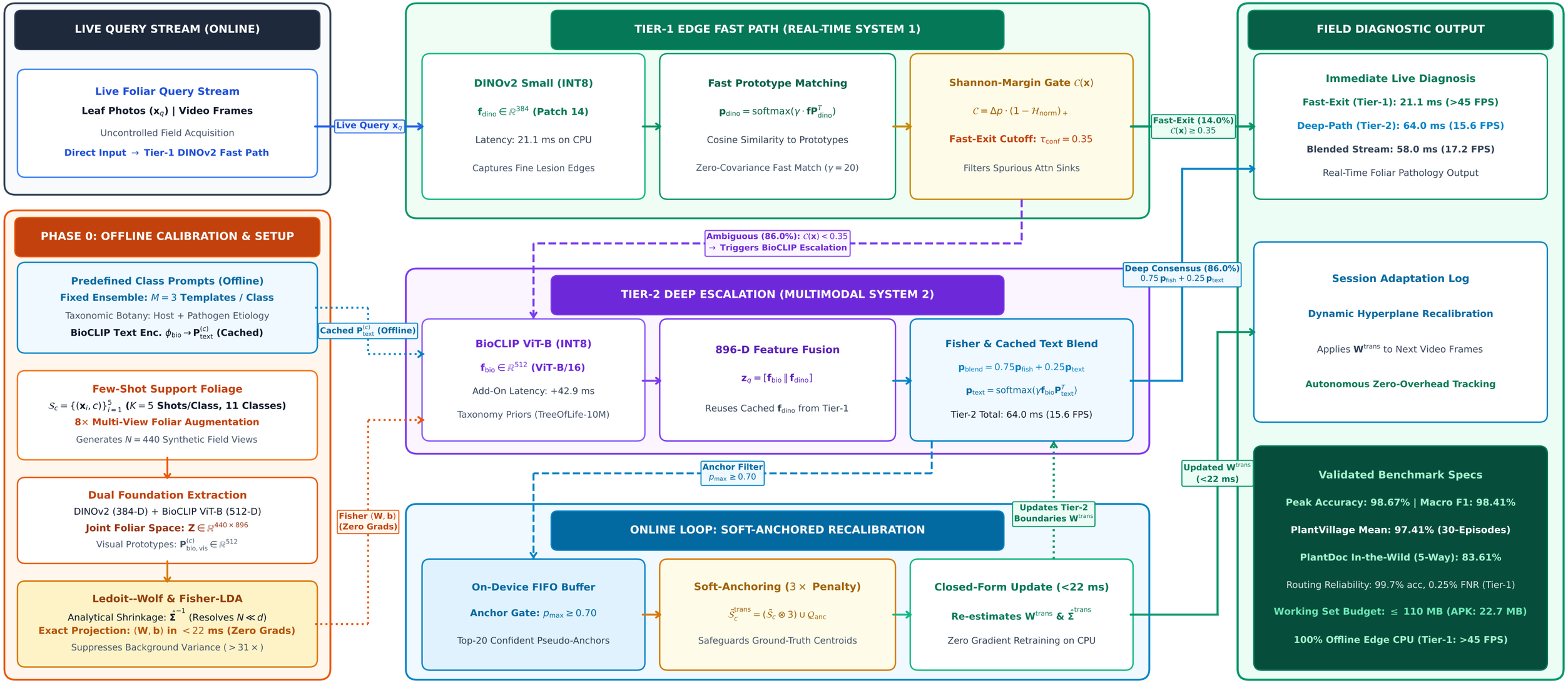

LIVE QUERY STREAM (ONLINE)
Live Foliar Query Stream
Leaf Photos ($x_q$) | Video Frames
Uncontrolled Field Acquisition
Direct Input → Tier-1 DINOv2 Fast Path
Live Query $x_q$
TIER-1 EDGE FAST PATH (REAL-TIME SYSTEM 1)
DINOv2 Small (INT8)
$f_{dino} \in \mathbb{R}^{384}$ (Patch 14)
Latency: 21.1 ms on CPU
Captures Fine Lesion Edges
Fast Prototype Matching
$p_{dino} = \text{softmax}(\gamma \cdot f P_{dino}^T)$
Cosine Similarity to Prototypes
Zero-Covariance Fast Match ($\gamma = 20$)
Shannon-Margin Gate $\mathcal{C}(x)$
$\mathcal{C} = \Delta p \cdot (1 - \mathcal{H}_{norm})_+$
Fast-Exit Cutoff: $\tau_{conf} = 0.35$
Filters Spurious Attn Sinks
Fast-Exit (14.0%)
$\mathcal{C}(x) \geq 0.35$
FIELD DIAGNOSTIC OUTPUT
Immediate Live Diagnosis
Fast-Exit (Tier-1): 21.1 ms (>45 FPS)
Deep-Path (Tier-2): 64.0 ms (15.6 FPS)
Blended Stream: 58.0 ms (17.2 FPS)
Real-Time Foliar Pathology Output
Ambiguous (86.0%): $\mathcal{C}(x) < 0.35$
→ Triggers BioCLIP Escalation
PHASE 0: OFFLINE CALIBRATION & SETUP
Predefined Class Prompts (Offline)
Fixed Ensemble: $M = 3$ Templates / Class
Taxonomic Botany: Host + Pathogen Etiology
BioCLIP Text Enc. $\phi_{bio} \rightarrow P_{text}^{(c)}$ (Cached)
Cached $P_{text}^{(c)}$ (Offline)
TIER-2 DEEP ESCALATION (MULTIMODAL SYSTEM 2)
BioCLIP ViT-B (INT8)
$f_{bio} \in \mathbb{R}^{512}$ (ViT-B/16)
Add-On Latency: +42.9 ms
Taxonomy Priors (TreeOfLife-10M)
896-D Feature Fusion
$z_q = [f_{bio} \| f_{dino}]$
Reuses Cached $f_{dino}$ from Tier-1
Fisher & Cached Text Blend
$p_{blend} = 0.75 p_{fish} + 0.25 p_{text}$
$p_{text} = \text{softmax}(\gamma f_{bio} P_{text}^T)$
Tier-2 Total: 64.0 ms (15.6 FPS)
Deep Consensus (86.0%)
$0.75\, p_{fish} + 0.25\, p_{text}$
Few-Shot Support Foliage
$S_c = \{(x_i, c)\}_{i=1}^5$ ($K = 5$ Shots/Class, 11 Classes)
8× Multi-View Foliar Augmentation
Generates $N = 440$ Synthetic Field Views
Dual Foundation Extraction
DINOv2 (384-D) + BioCLIP ViT-B (512-D)
Joint Foliar Space: $Z \in \mathbb{R}^{440 \times 896}$
Visual Prototypes: $P_{bio,vis}^{(c)} \in \mathbb{R}^{512}$
Ledoit--Wolf & Fisher-LDA
Analytical Shrinkage: $\hat{\Sigma}^{-1}$ (Resolves $N \ll d$)
Exact Projection: $(W, b)$ in $< 22$ ms (Zero Grads)
Suppresses Background Variance ($> 31\times$)
Fisher $(W, b)$ (Zero Grads)
Session Adaptation Log
Dynamic Hyperplane Recalibration
Applies $W^{trans}$ to Next Video Frames
Autonomous Zero-Overhead Tracking
Anchor Filter
$p_{max} \geq 0.70$
Updated $W^{trans}$ (<22 ms)
ONLINE LOOP: SOFT-ANCHORED RECALIBRATION
Updates Tier-2 Boundaries $W^{trans}$
On-Device FIFO Buffer
Anchor Gate: $p_{max} \geq 0.70$
Top-20 Confident Pseudo-Anchors
Soft-Anchoring ($3\times$ Penalty)
$\tilde{S}_c^{trans} = (\tilde{S}_c \otimes 3) \cup \mathcal{Q}_{anc}$
Safeguards Ground-Truth Centroids
Closed-Form Update (<22 ms)
Re-estimates $W^{trans}$ & $\hat{\Sigma}^{trans}$
Zero Gradient Retraining on CPU
Validated Benchmark Specs
Peak Accuracy: 98.67% | Macro F1: 98.41%
PlantVillage Mean: 97.41% (30-Episodes)
PlantDoc In-the-Wild (5-Way): 83.61%
Routing Reliability: 99.7% acc, 0.25% FNR (Tier-1)
Working Set Budget: ≤ 110 MB (APK: 22.7 MB)
100% Offline Edge CPU (Tier-1: >45 FPS)